Mass-balance dosimetry for in vitro PFAS testing: a source-verified reassessment of two models against three uptake datasets

Working manuscript. Analysis, figures and tables are reproducible from `tk_review/scripts/`; see *Data and code availability*.

Abstract

Background. Quantitative in vitro to in vivo extrapolation (QIVIVE) for per- and polyfluoroalkyl substances (PFAS) requires knowing how much of an administered dose actually reaches cells in a bioassay. Nominal medium concentrations overstate it, sometimes by three orders of magnitude, because PFAS bind extensively to serum albumin and sorb to vessel walls. Equilibrium mass-balance models (IV-MBM) were developed to recover the freely dissolved and cellular concentrations from first principles.

Objective. To reassess two published IV-MBM formulations — the single-descriptor IV-MBM EQP v2.0 of Armitage et al. and the concentration-dependent balance of Qin et al. — against the three published datasets reporting PFAS cellular uptake in human liver cell lines, with all model parameters and all exposure conditions verified against primary sources.

Approach. Both models were re-implemented from their published equations and supplementary parameterisations. The three uptake datasets (Rosenmai et al.; Amstutz et al.; Fragki et al.) were harmonised onto the fraction-of-dose axis. Exposure conditions were read from the primary methods sections rather than inherited from derived descriptions. Model residuals were decomposed by study, compound, dose and chain length, and each candidate parameter was replaced with the best available independent measurement.

Principal findings. (i) Reading the primary methods revealed that one of the three assays was dosed in 2% rather than 10% serum, a five-fold error in the dominant sorptive phase that is not apparent from secondary descriptions; correcting it degraded the apparent fit of the dataset that had previously looked best. (ii) On corrected media both models over-predict cellular uptake — geometric-mean bias 3.9× (Armitage) and 7.5× (Qin) over 29 observations. (iii) Every substitution of a best-available measured parameter made the fit worse, in each case because the measurement moved the balance in the direction the residual did not need. (iv) The residual separates into a level error and a chain-length error with different causes: the chain-length drift is governed by the ratio of the cell to the albumin distribution ratio, not by either term alone. (v) The same formalism applied to human serum reproduces measured whole-serum partitioning for 26 PFAS with no fitted parameter, establishing that the accounting is sound and localising the failure to the in vitro system. (vi) All three assays operate 5,500–27,700× above the physiological ligand:protein ratio, confirming and quantifying the regime concern raised by Qin et al.

Implication. Improved bookkeeping within the well yields a better description of the experiment but does not by itself yield a better oral equivalent dose, because the experiments occupy a binding regime that does not occur in vivo. We give specific reporting requirements that would make such assays usable for QIVIVE.

Keywords: PFAS; in vitro dosimetry; mass balance; QIVIVE; serum albumin; ligand:protein ratio; freely dissolved concentration

1. Introduction

1.1 The nominal concentration problem

In vitro bioactivity data are reported almost universally on a nominal basis: the concentration weighed into the medium. For chemicals that distribute appreciably out of the aqueous phase, that number is not the exposure the cells experienced. Chemical partitions into serum lipids and proteins, sorbs to the plastic of the well, is taken up by the cells themselves,

and — for volatiles — escapes to the headspace. The freely dissolved concentration that drives uptake can be orders of magnitude below nominal [1,12].

This matters because the purpose of the measurement is extrapolation. If an effect concentration measured in a dish is to be compared with a blood concentration in a person, the two must be expressed on comparable bases. A bioassay medium may contain anywhere from 0 to 20% serum; blood is blood. Using a nominal effect concentration as a direct surrogate for a blood concentration embeds a bias whose size depends on hydrophobicity and on assay composition [1].

Equilibrium mass-balance models were developed to close this gap [1,2,12,13,16]. They are now sufficiently mature that several independent implementations exist, and a comparison of their features has been published [17, cited via 1].

1.2 Why PFAS are a difficult case

The spLFER-based mass-balance approach was calibrated on neutral, hydrophobic organics, for which a single descriptor — the octanol–water partition ratio — predicts partitioning to most biological phases well. PFAS violate the premises of that approach in three distinct ways.

First, they are surfactants: a perfluorinated tail with an ionisable head group. Their behaviour at interfaces and in mixed phases is not well captured by a bulk hydrophobicity descriptor, and the descriptor itself is difficult to measure for them.

Second, they are essentially fully anionic at physiological pH. Ionisation is handled in these models by scaling the neutral-form partition ratio by a phase-specific penalty. For serum albumin, Armitage et al. set that penalty to zero for acids — the anion is assumed to bind albumin exactly as well as the neutral form would (ref. [1], SI Table S1). This is a modelling decision rather than a measurement, and it is consequential: it is a major reason albumin ends up carrying the great majority of the chemical in these calculations.

Third, PFAS bind albumin at discrete sites rather than dissolving into it diffusely. Site binding saturates. Qin et al. [3] measured PFAS–albumin isotherms over five orders of magnitude and found the binding strongly concentration-dependent, which is the basis of their model and the reason the linearity assumption common to spLFER approaches cannot simply be carried over.

1.3 Aims

This reassessment addresses four questions:

1. How well do the two models predict PFAS cellular uptake when both are implemented from their own published parameterisations?
2. Are the available uptake data mutually consistent enough to adjudicate between models?
3. Where does the residual lie, and can it be attributed to a parameter?
4. What do the answers imply for the use of these assays in QIVIVE?

We stress at the outset that question 4 was raised, and answered in principle, by Qin et al. [3], who hypothesised that concentration-dependent protein binding causes the binding regime in bioassays to differ substantially from that in plasma, with consequences for QIVIVE. What follows quantifies that gap for the specific uptake datasets used to parameterise and test these models, and examines what it leaves achievable.

2. The mass-balance formalism

2.1 Governing relation

The well is treated as a set of phases at equilibrium. Conservation of the administered amount, with each phase concentration expressed relative to the freely dissolved concentration by a distribution ratio D, gives

$$n_{total} = C_{free} \cdot (V_w + \Sigma_i D_i V_i)$$

so that $C_{free} = n_{total} / S$, where $S \equiv V_w + \Sigma D_i V_i$ is the total sorptive capacity of the well in units of water-equivalent volume. The fraction of the administered dose residing in cells follows as $x_{cell} = D_{cell} V_{cell} / S$.

Models differ only in how the D_i are obtained. The accounting is not in dispute.

2.2 Assumptions

The derivation requires: equilibrium; no degradation, metabolism or active transport; no losses; additive, non-interacting phases; and — the load-bearing one — linear sorption, i.e. constant activity coefficients, which holds only while the chemical is dilute with respect to every sorptive phase.

For PFAS in these assays, equilibrium is defensible: Fragki et al. [7] report cellular steady state within approximately 1 h against a 24 h exposure. Metabolic stability is defensible for perfluoroalkyl acids, though not for fluorotelomer alcohols. Active transport is assumed absent and cannot be checked. The linearity assumption is the one that fails outright, and §7 quantifies by how much.

2.3 Choice of output metric

Two metrics are available: the cellular concentration and the fraction of the dose in cells. The published uptake data are reported as amount per mass of cell protein, so converting them to a concentration requires a cell volume, while converting them to a fraction requires only protein per cell.

It should be noted — because it is easy to get wrong — that this does *not* make the fraction axis immune to cell-volume uncertainty. The predicted fraction is proportional to cell volume even though the measured fraction is not, with the result that the ratio of predicted to measured scales with assumed cell volume on *either* axis. What the fraction axis does buy is that the three datasets' *measurements* become mutually comparable without any cell volume being assumed. We use it for that reason.

3. The two models

3.1 IV-MBM EQP v2.0 (Armitage et al.)

The model predicts all required distribution ratios from the neutral-form octanol–water partition ratio via sorbent-phase-specific linear free energy relationships (spLFERs), with ionisation handled by phase-specific scaling factors [1]. The cell is represented as a proximate composition — storage lipid, membrane lipid, bulk protein and water — each contributing in proportion to its volume fraction. Plastic sorption is included via one of two published spLFERs.

We verified every parameter used here against the published supplementary material: both spLFERs (SI §S1.3), all four ionisation scaling factors (SI Table S1), and the cell-line compositions (SI Table S5). All agree exactly.

One parameter could not be verified. How bulk cell protein is converted into sorptive capacity is stated nowhere in the main text or the supplementary material; §S1 covers anionic phospholipids, lysosomal sequestration and the spLFERs, then defers to ref. [2] for temperature and ionic-strength adjustments only. Nor can ref. [2] contain it, since bulk protein is listed among the phases that v2.0 *added*. The implementation used here therefore adopts the convention in EPA's `httk` package, under which 3.5% of bulk protein is treated as octanol-equivalent. **Findings below that turn on this constant are findings about that implementation.** We return to this in §9.3.

The consequence of a small protein term is structural: with protein contributing 0.1–0.2% of the cell's sorptive capacity, the modelled cell is 79% water for the shortest chain tested and 99.7% membrane lipid for the longest. The cell is, in effect, a lipid droplet.

3.2 The concentration-dependent balance (Qin et al.)

Qin et al. [3] measured PFAS binding to bovine serum albumin, *gamma*-globulin, lipid and cells by C18 solid-phase microextraction over five orders of magnitude, and represent albumin binding as a saturable term plus a linear non-specific term. Their cell comprises protein and lipid with no aqueous phase, and the protein term is a measured non-specific protein–water ratio rather than an octanol surrogate. Their cell is 92–95% protein at every chain length.

The saturable albumin term can be derived from mass action: for n independent sites, the site concentration within pure albumin is $n \cdot rho / M$. Using the albumin density and molecular weight implied by their own reported values, this evaluates to $n \times 20{,}523$ µM, which reproduces their published site-capacity column to within 0.2% and implies $n = 0.45$ for PFBA

and 1.24 for PFOA. In the trace limit the term collapses to a constant, recovering the standard bridge between a binding constant and a partition coefficient.

Parameters were measured for four compounds (PFBA, PFOA, PFHxS, PFOS). Values for PFHxA and PFHpA are interpolated between the two carboxylate anchors and PFNA is extrapolated beyond both; only the lipid term has a published interpolation rule.

3.3 Structural comparison

Table 1. Structural differences between the two models.

	IV-MBM EQP v2.0	**Qin et al.**
source of *D*	predicted from log KOW	measured, then interpolated
albumin	linear, piecewise spLFER	saturable (site binding + non-specific)
albumin anion penalty	0.0 (acids)	not applicable; measured on the anion
cell phases	storage lipid, membrane lipid, 3.5%·protein, water	protein, lipid; no water
cell composition at short chain	~79% water	~92% protein
cell composition at long chain	~99.7% membrane lipid	~95% protein
dose dependence	linear by construction	non-linear
chemical coverage	any, from one descriptor	four measured compounds

The two models hold opposite views of what a cell is. Their predicted uptake nonetheless agrees to within a factor of ~2.5 in many cases, because albumin dominates the denominator in both and masks the difference. Agreement between them should not be read as corroboration of either cell representation.

3.4 The wider landscape, and why two models

Several equilibrium and kinetic distribution models for in vitro systems have been published; Proença et al. [17] review and compare them. They differ along three design axes that matter for PFAS:

Table 1b. Design axes across published in vitro distribution approaches.

	source of partitioning	**albumin treatment**	**cell representation**
Armitage et al. [1,2]	spLFERs from log KOW	linear, piecewise	lipid + 3.5%·protein + water
Escher-school balances [12,14,15]	**measured** biopartitioning	linear	lipid + protein, measured
Qin et al. [3]	measured, PFAS-specific	**saturable**	protein + lipid
Stadnicka-Michalak et al. [18]	regressions on log KOW	linear	kinetic, not equilibrium
VCBA / PBK-coupled [19]	varies	linear	kinetic, coupled to PBK

Two of these — the kinetic approaches — answer a different question, since the PFAS uptake data here reach steady state within ~1 h against a 24 h exposure and an equilibrium treatment is adequate. Of the equilibrium models, the two tested here occupy opposite corners of the first two axes, which is why they were selected. The third corner — measured biopartitioning with a *linear* albumin term, the Escher-school design — can be constructed from published PFAS measurements without reimplementing anyone's code, and §6.7 does so.

4. The empirical basis

4.1 Three datasets

Three studies report PFAS cellular uptake in human liver cell lines on a fraction-of-dose basis: Rosenmai et al. [5] (HepG2, 14 PFAS, three doses), Amstutz et al. [6] (HepG2, 13 PFAS, one dose) and Fragki et al. [7] (differentiated HepaRG, four PFAS, five doses). Twenty-nine observations overlap the five compounds for which both models can be parameterised.

4.2 Exposure conditions, read from source

Table 2. Exposure conditions as stated in the primary methods sections. Entries in bold differ from what a reader would infer from the cell-culture section alone.

	Rosenmai [5]	**Amstutz [6]**	**Fragki [7]**
cell line	HepG2	HepG2	differentiated HepaRG
growth medium serum	10% FBS	10% FBS	10% FBS
exposure medium serum	10% FBS	10% FBS	**2% FBS**
co-solvent	**3% ethanol**	0.1% DMSO	0.5% DMSO
plate format	12-well, 3 wells pooled	96-well	24-well
volume dosed -> after t=0 sample	not stated	250 -> 200 µL	550 -> 500 µL
wash	5× ice-cold PBS	5× DPBS	5× DPBS
uptake normalised to	nominal	**recovered**	nominal

Two entries require comment. Fragki et al. state that exposure was conducted in "assay medium (growth medium containing 2% FBS)"; the 10% figure belongs to the growth and differentiation medium. Because serum fraction multiplies the volume of the dominant sorptive phase, this is a five-fold difference in the parameter to which the balance is most sensitive, and it is not recoverable from descriptions that quote the culture conditions. Second, Amstutz et al. normalise their intracellular fraction to *recovered* rather than administered mass, so their denominator excludes material lost to the vessel; their values are not strictly on the same axis as the other two.

Rosenmai et al. do not state a working volume; 1 mL per well is assumed, as in all prior analyses of this dataset known to us.

4.3 Inter-study consistency

The three datasets disagree with one another before any model is involved. At 100 µM, cellular uptake of PFNA is reported as 0.31% [5], 0.30% [6] and 4.39% [7] — a 14-fold spread. For PFHxA at the same dose the two HepG2 studies differ by 3.6-fold (0.14% [5] vs 0.039% [6]) despite using the same cell line and nominally similar protocols; part of this is attributable to the different normalisation noted above.

Any model-performance statement must be read against this. A model disagreeing with these data by a factor of four is disagreeing with a target that is itself uncertain by a comparable factor.

5. Methods of this reassessment

Both models were implemented in Python from their published equations and parameterisations (`mbm_published_equations.py`). Armitage parameters were taken from ref. [1] and its supplementary material; Qin parameters from ref. [3]. Where the Armitage implementation required a parameter not present in the published record (cell protein sorption, §3.1), the `httk` convention was used and is flagged throughout.

Assay geometry, cell number, serum fraction and dose series were taken from the primary publications (Table 2). Measured uptake was placed on the fraction-of-nominal-dose axis. Model performance is reported as geometric-mean bias (predicted ÷ measured) and geometric-mean absolute error.

Residuals were decomposed by study, by compound, by dose and against perfluorocarbon chain length. Four parameters were then replaced in turn with the best available independent measurement: albumin partitioning from Fischer et al. [9]; cell partitioning from the measured values in ref. [3]; cell protein sorption from ref. [4]; and cell volume from an independent HepG2 determination.

6. Results

6.1 Effect of the medium correction

Implementing Fragki et al. at the serum fraction stated in their methods rather than the growth-medium value changes that dataset's geometric-mean bias from 1.54× to 14.92× (Qin) and from 2.48× to 10.80× (Armitage). Pooled across all 29 observations, bias moves from 3.43× to 7.52× (Qin) and 2.33× to 3.87× (Armitage).

The direction is worth stating plainly: the dataset that had appeared to be the models' best case was the one being supplied with five times more albumin than it contained. Correcting an error degraded the apparent fit. This is the expected behaviour when a model is accidentally compensated, and it is a cautionary result for any benchmarking exercise that inherits assay conditions from secondary sources.

The Armitage response (4.4×) matches the denominator arithmetic, since its albumin term is linear. The Qin response (9.7×) exceeds it, because a saturable sink amplifies an error in its own size: reducing albumin raises the free concentration, which raises site occupancy, which lowers the apparent distribution ratio, which reduces capacity further. The amplification is largest at intermediate doses, where the 10% and 2% wells fall on opposite sides of saturation, and vanishes at the top dose where both are saturated.

6.2 Overall performance

Table 3. Model performance on corrected media, 29 observations.

	geometric-mean bias	**geometric-mean **	**error**
IV-MBM EQP v2.0	**3.87×**	**4.02×**	
Qin et al.	7.52×	7.52×	

by study	**Armitage**	**Qin**
Rosenmai [5] (n=15)	1.97×	3.72×
Amstutz [6] (n=4)	3.77×	18.95×
Fragki [7] (n=10)	10.80×	14.92×

Both models over-predict throughout. The error is not uniform across compounds: for PFBA–PFHpA, which only the two 10% FBS studies measure, Armitage bias is 1.2–1.6×, while PFOA and PFNA — the compounds Fragki contributes — carry almost the entire residual.

6.3 Dose dependence

Armitage is linear by construction and its fitted dose exponent is pinned at 1.00. Measured exponents run 0.85–1.42 and rise with chain length. Qin reproduces that ordering, overshooting by 0.30–0.71 on the Rosenmai series and by 0.07–0.20 on the Fragki series. This is the one respect in which the additional machinery of the saturable model buys something the linear model structurally cannot deliver, and it is also the one respect in which the medium correction *improves* Qin: the Fragki exponents fall from 1.71 and 1.66 to 1.49 and 1.62 against a measured 1.42, indicating that part of the curvature previously attributed to chemistry was an artefact of an oversized albumin sink.

6.4 Parameter substitutions

Table 4. Effect of replacing each parameter with the best available independent measurement. Bias is the geometric mean over 29 observations.

substitution	**direction of the measurement**	**Armitage**	**Qin**
published models	—	3.87×	7.52×
albumin partitioning [9]	**below** both models	8.08×	7.75×
cell partitioning (measured, [3])	mixed	2.15×	4.61×
cell protein sorption [4]	**above** both models	9.81×	12.24×
cell volume (independent HepG2)	—	**1.84×**	3.71×

Three of the four substitutions worsen the fit, and they do so for a consistent reason: each measurement moves the balance in the direction the residual does not require. A weaker albumin reduces competition and sends more chemical

into the cell; a stronger cell-protein affinity enlarges the cell sink and does the same. Both models already over-predict.

The cell-partitioning substitution improves the aggregate but at the cost of destroying the chain-length trend, over-predicting PFBA by 19.7× (Qin) and 23.6× (Armitage) while landing within a factor of 1.4 at PFNA. Aggregate improvement achieved by averaging a large miss at one end of a series against a good fit at the other is not evidence of a better parameter.

Only cell volume improves the fit without that penalty, and it is the one parameter none of the three studies measured. It is also the only parameter that rescales every compound by the same factor, which is what an offset independent of chain length requires. On corrected media it accounts for roughly half the residual in log terms and leaves a 1.8–3.7× over-prediction standing.

This is the central negative result of the reassessment: a parameter can be the most influential term in a model and not be the term that is wrong. Albumin carries 94–97% of the sorptive capacity for the longer chains, which makes it the obvious suspect and the obvious thing to tune; the best available measurement of it moves the prediction further from the data.

6.5 Chain-length structure of the residual

Armitage's over-prediction rises monotonically with perfluorocarbon chain length; Qin's does not. Within the Rosenmai dataset alone — one laboratory, one cell line, one medium, one vehicle — the Armitage residual climbs 0.86× to 9.33× between PFHxA and PFNA. Qin shows no trend within any single study.

The cause is not either distribution ratio alone. Where albumin dominates the denominator, the predicted fraction in cells varies as the *ratio* $D_{cell}/D_{albumin}$, so the residual's chain-length slope is the error in the slope of that ratio.

Table 5. Chain-length dependence over nCF 5–8, decades per CF_2.

	albumin	cell	ratio	residual, observed
IV-MBM EQP v2.0	+0.386	+0.769	**+0.383**	**+0.345**
Qin et al.	+0.547	+0.515	**-0.032**	**-0.074**
measured albumin [9]	**+0.540**	—	—	—

The ratio slope predicts the residual slope to within the uncertainty of a four-point fit. Armitage's albumin spLFER rises more slowly with chain length than the measurement, because the piecewise form flattens above log KOW 4.5 — a feature calibrated on neutral organics and pharmaceuticals, where a binding cavity of finite depth does plateau. Its cell term rises faster than a constant ratio would require, because a cell represented as lipid gains affinity with chain length faster than albumin does. The two errors compound, contributing roughly 40% and 60% of the gap respectively.

Qin's albumin slope sits within 0.007 of the measured value and his ratio is flat. **The two models therefore fail in different ways: Qin has the chain-length behaviour approximately right and the level wrong; Armitage has the level closer and the chain-length behaviour wrong.**

Independent estimates of the cell distribution ratio's chain-length slope span +0.177 (cell homogenate, [3]) to +0.782 (the Armitage parameterisation), with the uptake-implied value (+0.407) and whole-tissue unbound fractions [11] (+0.46 to +0.53) in between. The quantity is bounded, not known.

6.6 What does not explain the residual

Two candidate artefacts were tested and rejected.

Wash losses. All three studies wash five times with cold buffer, so it cannot be a between-study variable; and the direction is wrong, since washing should preferentially strip the shortest, most weakly bound chains, whereas over-prediction rises with chain length.

Cytotoxicity at the long-chain end. Reduced protein content at high dose is documented in two of the three studies, and Fragki et al. excluded PFNA at 100 µM from their own analysis on those grounds. If cell loss were suppressing measured uptake, the chain-length trend would be strongest at the highest dose. It is strongest at the lowest (+0.194 at 10 µM, +0.099 at 100 µM).

Co-solvent. The three assays differ 30-fold in vehicle strength (Table 2). A co-solvent lowers every water-to-phase distribution ratio by approximately the same factor, and the predicted fraction is a ratio of sorptive terms, so the effect cancels except against the aqueous term. It is therefore material only where water carries a substantial share of the capacity — PFBA, where it changes the residual from 1.42× to 1.11×, and negligible elsewhere (PFNA, 9.33× to 9.31×). The 30-fold difference in vehicle is, counter-intuitively, of almost no consequence for this comparison.

6.7 A third parameterisation: measured biopartitioning

The design decision separating the Escher-school balances [12,14,15] from IV-MBM EQP v2.0 is the use of measured membrane–water and protein–water partitioning in place of single-descriptor regressions, retaining a linear albumin term. Armitage et al. report that this substitution substantially improves their own model's performance for ionisable organics [1].

That variant can now be built for PFAS from published measurements: albumin from Fischer et al. [9] and phospholipid membrane–water coefficients from Chen et al. [10], substituted into the Armitage cell composition and plastic term. PFBA is excluded (no measured albumin value), so comparisons are over nCF 5–8 and are directly comparable with the corresponding columns in §6.5.

Both measured series sit below the spLFER predictions — albumin by 2–7× and membrane lipid by 8–90×, with the membrane discrepancy growing systematically with chain length.

Table 6. Performance over nCF 5–8.

parameterisation	bias	slope (decades/CF_2)
Armitage, as published	4.35×	+0.316
+ measured albumin only	10.12×	+0.194
+ measured membrane only	0.16×	-0.049
+ both measured (Escher-school design)	**0.40×**	**-0.156**
Qin, as published (saturable albumin)	8.59×	-0.043

Two observations follow, and they point in different directions.

First, **this is the only substitution attempted in this work that moves the level toward unity** rather than away from it, and it does so decisively — from 4.35× over-prediction to 0.40×, i.e. 2.5× under. The term responsible is the membrane-lipid spLFER, not albumin and not cell protein. Given that §6.4 found every other best-available measurement worsening the fit, this deserves emphasis: it identifies the membrane-lipid term as a candidate location for the level error that the earlier substitutions had not reached.

Second, and more robustly, **the chain-length drift is not removed by measured parameters alone**. It falls from +0.316 to -0.156 but does not vanish, and it is the *saturable* albumin treatment that flattens it (-0.043), at the cost of the worst level of the four.

An important caveat attaches to the first observation. The basis on which Chen et al. report KMW is defined in a supplementary equation not available to us, so the absolute offset between the measured and predicted membrane series may include a basis mismatch. The *slope* difference — the measured series rising ~0.44 decades per CF_2 against ~0.78 for the spLFER — is invariant to any constant basis error and is the part we would defend. The level shift should be treated as provisional pending that definition.

The wider implication for the landscape is that the two failure modes documented here map onto two independent design axes. Measured biopartitioning addresses the level; a saturable albumin term addresses the chain-length behaviour. No published model currently occupies the corner that combines both with a protein-realistic cell.

7. The ligand:protein regime

Qin et al. [3] hypothesised that concentration-dependent protein binding makes the binding regime in bioassays differ substantially from that in plasma, with consequences for QIVIVE. Figure 1 quantifies that gap for the three uptake

datasets.

Albumin offers approximately one high-affinity site per molecule for these compounds (§3.2). The two 10% FBS assays contain 36.1 µM albumin and the 2% FBS assay 7.2 µM, against nominal doses of 6.25–100 µM. The resulting nominal ligand:albumin ratios are 0.28–2.77 and 0.87–13.85 mol/mol respectively. Fischer et al. [9] worked at <= 0.004 mol/mol and estimate the ratio in human serum at ~ 0.0005. **The assays therefore operate 5,500–27,700× above the physiological regime.**

The consequence for the models is direct. Solved self-consistently within the wells, albumin site occupancy runs from 20% to 99% across the dose series, and the apparent albumin distribution ratio falls to between 0.11 and 0.07 of its trace-limit value — a sevenfold change in a quantity that a linear model treats as a constant, within a single experiment (Fig. 1B).

This is why the Qin formulation requires its saturable term and why a linear spLFER treatment cannot be correct here, whatever its merits for the neutral organics it was calibrated on.

It also bounds what the exercise can deliver. Even an exactly correct model of these wells would describe a binding regime that does not occur in vivo. Improving the in vitro mass balance improves the description of the experiment; it does not, on its own, improve the extrapolation, because the experiment is in the wrong regime.

(Figure 1: `figures/fig21_mbm_regime_gap.png`*. A, nominal ligand:albumin ratio against dose for the three assays with the physiological band marked; B, apparent albumin distribution ratio relative to its trace limit as a function of freely dissolved concentration, with the positions occupied by the three assays marked.)*

8. A positive control: the same formalism in serum

The results above could be read as indicting the mass-balance approach itself. A control argues otherwise.

Fischer et al. [9] measured, by one method in one laboratory, PFAS distribution ratios to purified human serum albumin, to *gamma*-globulin, and to whole human serum, at a ligand:protein ratio of <= 0.004. The components and the whole matrix are separate experiments, so predicting the latter from the former is an out-of-sample test of the identical accounting used above.

For the 26 PFAS with all three measurements, the balance evaluated at population-average serum composition and with no fitted parameter gives a geometric-mean bias of 1.00× and a geometric-mean absolute error of 1.55× (RMSE 0.262 log units; R^2 = 0.68 about the 1:1 line) for a quantity spanning nearly two orders of magnitude. Allowing the two protein concentrations to float returns 39.5 and 38.6 g/L against population values of 41 and 31 g/L and does not reduce the RMSE at all — the model is not being supported by its parameters. Globulin earns its term only above the ratio threshold the authors themselves propose. Adding a phospholipid sink improves nothing.

The accounting is therefore sound, and the failure is specific to the in vitro system — which has more phases, a vessel-wall term that is poorly constrained, cell volumes nobody measured, and a ligand:protein regime in which the central linearity assumption does not hold.

(Figure 3: `figures/fig19_fischer_serum_mbm.png`*.)*

9. Discussion

9.1 Implications for QIVIVE

Three findings bear on the use of these assays for reverse dosimetry.

First, the models over-predict cellular uptake by factors of 2–10 on data whose own inter-study spread is a factor of 14. Neither model is currently accurate enough to carry a QIVIVE calculation without an uncertainty statement of comparable size.

Second, the residual is not attributable to any parameter for which a better measurement exists. Substituting the best available value for albumin, for cell partitioning and for cell protein each worsens the fit. The only parameter that improves it is the one nobody measured.

Third, and most consequentially, the regime gap of §7 means that improving the in vitro balance does not translate directly into a better oral equivalent dose. Qin et al. [3] reached the same conclusion from the opposite direction, showing that free-concentration-based QIVIVE ratios exceed nominal-based ones precisely because of protein-content differences between blood and bioassays. Our contribution is to show that the gap is large enough — three to four orders of magnitude in ligand:protein ratio — that the saturation state of albumin differs qualitatively, not merely quantitatively, between the two settings.

9.2 Reporting recommendations

Several of the difficulties encountered here are avoidable at no experimental cost.

1. **Report the exposure medium composition, not the culture medium composition.** The single most consequential error in this work arose from a five-fold difference between the two in one study, stated in one sentence of its methods and absent from derived descriptions.
2. **Report the working volume.** One of the three studies does not, leaving a parameter that enters the balance linearly to be assumed.
3. **Report the cell volume, or cells per well together with protein per cell.** Cell volume is the only parameter that improved the fit here and the only one none of the studies measured.
4. **Report the ligand:protein molar ratio.** It determines whether a measured binding constant is in the trace regime and therefore whether it can be transferred to another setting at all.
5. **State the normalisation basis** — administered or recovered — since the two differ by whatever was lost to the vessel.
6. **Report the mass balance where it was measured.** Two of the three studies took a t = 0 medium sample explicitly for this purpose; neither reports the result. Doing so would constrain the vessel-wall term directly rather than by difference.

9.3 Limitations of this reassessment

The Armitage implementation depends on one parameter — the sorptive capacity assigned to bulk cell protein — that is not documented in the published record (§3.1). We used the `httk` convention. Findings in §6.4 and §6.5 that turn on the size of the cell protein term are therefore findings about that implementation; whether they apply to the authors' own tool cannot be decided from the papers, and would require the Excel/VBA implementation, which is available on request. The structural argument is unaffected: a cell whose protein term is negligible is represented as lipid, and a lipid representation carries too steep a chain-length dependence.

Qin's parameters exist for four compounds; two of the five modelled here are interpolated and one extrapolated. The extrapolated compound, PFNA, is among the worst-fitting. More seriously, **the full text of ref. [3] was not available to us**: the model was implemented from its published equation form and tabulated parameters. Any qualification the authors place on their own parameterisation that is not visible in the tables would not have been captured.

The measured-membrane result of §6.7 rests on a coefficient whose published basis is defined in a supplementary equation we could not obtain; the level shift it produces is provisional for that reason, as stated there.

The uptake datasets are small, mutually inconsistent, and partly not on a common normalisation basis. Rosenmai's working volume is assumed. Conclusions about absolute bias are correspondingly weaker than conclusions about *direction* and about *structure*, which is why we have emphasised the latter.

Finally, we have not evaluated the other published in vitro distribution models [17, cited via 1], and the comparison here should not be read as a ranking of the field.

10. Conclusions

1. Both mass-balance models over-predict PFAS cellular uptake in human liver cell lines, by 3.9× (IV-MBM EQP v2.0) and 7.5× (Qin et al.) in geometric mean, once exposure conditions are read from primary sources.
2. Exposure media differ between the three available datasets in ways that dominate the model comparison and are not recoverable from secondary descriptions. One assay was dosed at one-fifth the serum concentration generally assumed for it.
3. Every substitution of a best-available measured parameter worsens the fit. Albumin is the most influential term in the balance and is not the source of the error.
4. The residual separates into a level component and a chain-length component. The latter is governed by the ratio of cell to albumin distribution ratios; the two models get opposite halves of this right.
5. All three assays operate three to four orders of magnitude above the physiological ligand:protein ratio, with albumin between 20% and 99% saturated. The linearity assumption underlying spLFER-based balances does not hold in this regime.
6. The same formalism applied to human serum reproduces measured whole-serum partitioning for 26 PFAS with no fitted parameter. The accounting is sound; the in vitro system is the problem.
7. Consequently, better in vitro dosimetry yields a better description of the experiment, but not by itself a better oral equivalent dose. Closing that gap requires assays run nearer the physiological binding regime, not better models of assays that are not.

Data and code availability

model implementations	`tk_review/scripts/mbm_published_equations.py`
exposure conditions, source-verified	`tk_review/scripts/mbm_medium_composition.py`
chain-length decomposition	`tk_review/scripts/mbm_chain_length_residual.py`
parameter substitutions	`mbm_fischer_albumin.py`, `mbm_cell_term.py`, `mbm_cell_protein.py`
serum control	`tk_review/scripts/mbm_fischer_serum.py`
Figure 1	`tk_review/scripts/make_mbm_regime_figure.py`
Figure 2	`tk_review/scripts/make_mbm_correction_figure.py`
Figure 3	`tk_review/scripts/make_mbm_fischer_serum_figure.py`
primary sources held locally	`literature/fulltext/`

Extended derivations and the full approximation inventory are in `report/MBM_PRIMER.md`; a non-technical exposition is in `report/MBM_UNDERSTANDING.md`; the detailed working memo, including material not carried into this manuscript, is `report/MBM_Qin_vs_Armitage.md`.

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Reference 3 is one of the two models under assessment; its parameterisation was taken from tabulated values and its full text has not been read. This is stated among the limitations (§9.3).